

The effect of hypertonic saline resuscitation on bacterial translocation after hemorrhagic shock in rats

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*Translocation of enteric bacteria occurs in rats after hemorrhagic shock. A proposed mechanism involves intestinal mucosal injury by hypoperfusion. Recent work suggests that moderate hypovolemia causes gut arteriolar constriction, which is ameliorated by hypertonic saline resuscitation. Bacterial translocation should, therefore, be reduced when hypertonic saline (HS) is used as the resuscitative fluid. Seventy-eight Sprague-Dawley rats were anesthetized and subjected to 30 minutes of hemorrhagic shock (systolic blood pressure 30 to 50 mm Hg) through a modified Wigger's model. Resuscitation was performed with either shed blood (B), 3% HS + 1/2B (1:1), or with 7.5% HS + 1/2B (1:1). Spleen, liver, and mesenteric lymph nodes were sent for quantitative culture 24 hours later. Translocation occurred if enteric organisms were cultured from at least one organ. Statistical analysis used the Fisher exact test. Compared to autotransfusion, hemodilutional resuscitation from hemorrhagic shock with hypertonic saline resulted in a significant reduction in bacterial translocation (*p* values were 0.03 and 0.04 for 3% and 7.5% hypertonic saline, respectively). The reduction in translocation after hypertonic saline resuscitation may be the consequence of microcirculatory alterations preventing gut hypoperfusion. (SURGERY 1991;110:685-90.)*

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NUMEROUS ADVANCES IN THE CARE of patients who have suffered trauma have been made in recent years. Blood transfusion, antibiotic therapy, and rapid transport to specialized trauma centers have resulted in a significant reduction in deaths after traumatic insult. One problem that continues to plague the severely injured, however, is multiple organ failure syndrome. The cause of this sequential "shut-down" of the body's vital organs is unknown. A leading theory about the cause of this syndrome centers around bacteria normally residing in the bowel.¹ Ravin et al.² in the 1960s, were the first to im-

plicate the gastrointestinal tract as a source of occult sepsis preceding multiple organ failure. Systemic infection presumably occurs when viable gut organisms traverse the bowel mucosa and travel to more distant sites, a process known as bacterial translocation.¹ Clinical evidence has supported the hypothesis that post-traumatic systemic infection is attributable to enteric organisms.³

Bacterial translocation has been shown in rats after hemorrhagic shock.⁴⁻⁶ Histologic findings suggest that shock-induced mucosal damage and, therefore, disruption of the gut mucosal barrier provides the port of entry for the organisms.⁵ The mechanism of mucosal disruption is related to intestinal hypoperfusion resulting from both hypovolemia and splanchnic vasoconstriction. If vasoconstriction were reversed and blood flow restored to the gut before occurrence of significant mucosal damage, one would expect a reduction in bacterial translocation. Scalia et al.⁷ have shown that when rats are resuscitated from hemorrhagic shock with the vol-

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Table I. Summary of translocation by resuscitative group

Resuscitative group	+TR	-TR	%TR	<i>p</i> Value
B	8	17	32	
3% HS + 1/2B	1	20	4.7	0.03
7.5% HS + 1/2B	2	22	8.3	0.04

+TR, Number of animals showing bacterial translocation; -TR, animals without evidence of translocation; %TR, number of translocating animals per total number comprising the resuscitative group. *p* Values were calculated comparing translocation after resuscitation by autotransfusion versus hypertonic saline.

ume of blood lost, splanchnic arteriolar constriction persisted. If the animal was resuscitated with a hypertonic saline solution, however, vessels returned to their preshock diameter. Hypertonic saline resuscitation, by restoring both circulatory volume and normal splanchnic arteriolar caliber, should reduce mucosal damage and subsequent bacterial translocation.

MATERIAL AND METHODS

This study was approved by the Animal Care Committee of the Hektoen Institute for Medical Research, Cook County Hospital.

Seventy-seven male Sprague Dawley rats weighing 277 to 528 grams (average, 366 grams) were used by this study. Each rat was anesthetized with 50 mg/kg pentobarbital intraperitoneally. Once adequate anesthesia was attained, the animal was placed in a supine position. The abdomen and both groins were shaved, prepped, and draped in a sterile fashion.

An incision was made in the right groin, and the femoral vessels were exposed. Proximal and distal control of the femoral artery was achieved, and the vessel was cannulated with polyethylene tubing (PE-20; Fisher Scientific Co., Pittsburgh, Pa.). The distal end of the tubing was connected to two three-way stopcocks joined in tandem. Systolic blood pressure was recorded on a polygraph (TA 2000; Gould, Inc., Cleveland, Ohio) connected to a stopcock port through a transducer (P23XL; Gould, Inc., Medical Product Division, Oxnard, Calif.). A heparin-saline flush and blood-collecting reservoir (heparinized 10 cc syringe) occupied the remaining stopcock ports.

Once cannulation of the femoral artery was complete, blood was withdrawn into the collecting reservoir until a systolic blood pressure of 30 to 50 mm Hg was achieved. Hemorrhagic shock was maintained at this level for 30 minutes by either withdrawal or return of shed blood.

The rats were resuscitated on completion of the shock period with either shed blood ($n = 25$), 3% hypertonic saline ($n = 21$), or 7.5% hypertonic saline ($n = 24$). Animals resuscitated with shed blood alone received the full volume of blood extracted. Those resuscitated with hypertonic saline (either 3% or 7.5%) first received the selected hypertonic saline concentration in a volume equivalent to one half of the blood volume extracted. This was then followed by an equal volume of shed blood.

The catheter was removed, and the femoral artery was ligated on completion of resuscitation. The incision was closed, and the animal was returned to its cage where food and water were available ad libitum.

A control group of seven rats was subjected to a 30-minute period of sham shock. These animals were instrumented in the manner described previously. No blood was withdrawn nor was resuscitative fluid administered.

Twenty-four hours after hemorrhagic or sham shock, the animal was returned to the operating suite. General anesthesia was achieved once again with pentobarbital. The liver, spleen, and mesenteric lymph node complex were harvested through a midline incision. The rat was then killed by intracardiac potassium chloride injection.

Quantitative cultures were obtained on the harvested specimens. Serial dilutions were plated on both sheep blood and McConkey's agar. After 24 hours of incubation at 37° C under aerobic conditions, the cultures were interpreted by a microbiologist blinded to the mode of resuscitation. The occurrence of translocation was defined as growth of enteric bacteria from at least one organ.

Statistical analysis comparing translocation in the various resuscitative groups used the Fisher exact test. ANOVA was used to compare the average weights and total blood volume extracted for each group. *P* values < 0.05 were considered statistically significant.

RESULTS

The average weight of rats was calculated for each resuscitative group. These weights were then compared in an effort to establish that the groups were composed of animals of similar age. ANOVA revealed no statistically significant difference in the weights of the animals between groups ($p > 0.25$).

Severity of shock was determined through total blood volume extracted. The average percentage of total blood volume lost for the three groups experiencing hemorrhagic shock was 42%. ANOVA revealed no significant difference in this statistic between groups ($p > 0.25$).

Animals resuscitated with autologous blood had the highest rate of enteric bacterial translocation (32%). Comparatively, the incidence of translocation after resuscitation with 3% hypertonic saline (4.7%) was significantly reduced ($p = 0.03$). A similar phenomenon occurred with 7.5% hypertonic saline resuscitation, with 8.3% of the animals showing translocation ($p = 0.04$). No significant difference was noted in the frequency of translocation between the two concentrations of hypertonic saline ($p = 0.55$). Translocation results are summarized on Table I.

Although animals were not resuscitated to a predetermined physiologic endpoint, all displayed pink mucosal membranes, reestablished their preshock blood pressure, and survived for 24 hours after resuscitation. Additionally, most rats spontaneously micturated before leaving the operating suite.

The most common tissue exhibiting enteric translocation was the mesenteric lymph node, occurring in 61% of all translocating animals. The liver cultured enteric organisms in 33%, and the spleen provided evidence of translocation in 8%.

Escherichia coli was the organism most commonly cultured. Other translocating organisms included *Enterobacter cloacae*, *Enterococcus*, *Acinetobacter antratus*, *Morganella morganii*, *Pseudomonas maltophilia*, *Staphylococcus aureus*, *Providencia* and *Proteus* species. Organisms cultured, and the frequency of occurrence are shown on Table II.

None of the animals subjected to sham shock showed translocation. Death before organ harvest was less than 5% for all groups combined. These animals were not included in the study results.

DISCUSSION

The process of bacterial translocation is influenced by three factors: (1) physical disruption of the gut mucosal barrier, (2) impaired host immunity, and (3) enteric bacterial overgrowth.⁸ All three factors are probably present to a degree in the human who has suffered trauma. Experimental studies of translocation after hemorrhagic shock in rats, however, suggest that bacterial overgrowth is not a dominant feature in this model. Deitch et al.⁴ have shown that cecal bacterial content after 30 minutes of hemorrhagic shock is no different from that of rats subjected to sham shock. This certainly does not discount gram-negative overgrowth as a significant factor influencing translocation in humans. Administration of broad spectrum antibiotics, for instance, may well alter the population of enteric microorganisms enough to enhance translocation.

Table II. Frequency of enteric organisms cultured

Organism	Frequency (%)
<i>Escherichia coli</i>	42
<i>Enterobacter cloacae</i>	17
<i>Enterococcus species</i>	17
<i>Staphylococcus aureus</i>	17
<i>Acinetobacter antratus</i>	8
<i>Morganella morganii</i>	8
<i>Pseudomonas maltophilia</i>	8
<i>Providencia species</i>	8
<i>Proteus species</i>	8

Frequency is based on number of animals with cultures positive for the given organism per total number of animals showing translocation.

Immunologic alterations after traumatic insult are well known. Shortly after hemorrhagic shock, neutrophil chemotaxis and phagocytosis are diminished.⁹ Additionally, reduced opsonin levels¹⁰ and reticuloendothelial cell activity¹¹ are noted. Cell-mediated immunity is also affected as T-cell helper-to-suppressor ratios are reversed.¹² Such immunosuppression may predispose the patient who has suffered trauma to bacterial translocation and subsequent systemic infection. The immunologic effect of translocation, however, was not specifically addressed in this study.

The model of hemorrhagic shock-induced translocation used by this study focuses on the remaining mechanism of translocation—disruption of the gut mucosal barrier. Using a similar model, Deitch et al.⁵ have shown focal areas of ileal and cecal mucosal injury within 24 hours of the shock period. Enhanced permeability to tracers, such as horseradish peroxidase, provides further evidence of mucosal compromise.⁵

Our data suggests that mucosal injury subsequent to hemorrhagic shock may be directly related to tissue ischemia. The splanchnic vasoconstriction and hypovolemia of hemorrhagic shock produce a significant strain on barrier integrity. Even when circulatory volume is restored by autotransfusion, gut arteriolar constriction persists.⁷ If hypertonic saline is added to the resuscitative regimen, however, both volume and vessel diameter are restored. This phenomenon could have a direct effect on the occurrence of translocation. When rats were resuscitated from a relatively short period of hemorrhagic shock (that is, 30 minutes), translocation was significantly reduced compared to animals resuscitated by autotransfusion alone.

Several investigators have suggested that gut mucosal damage subsequent to hemorrhagic shock is primarily the consequence of reperfusion injury. The incidence of

shock-induced translocation is diminished when rats are pretreated with allopurinol.³ The mechanism by which this occurs is attributed to inhibition of xanthine oxidase. Retarding hydroxyl radical production by this enzyme prevents intestinal injury and, therefore, bacterial translocation.

The process of bacterial translocation after hemorrhagic shock is complex, involving numerous interacting variables. This is readily illustrated by the cause of gut mucosal disruption. Hypertonic saline is not known to inhibit xanthine oxidase nor to possess hydroxyl radical scavenger capabilities. It is likely, therefore, that hypertonic saline resuscitation protects mucosal integrity by restoring oxygen delivery as opposed to directly preventing reperfusion injury. It is possible that, through restoration of circulatory volume and preshock splanchnic arteriolar diameter, hypertonic saline retains gut barrier function, thus hindering enteric translocation subsequent to a short period of hemorrhagic shock.

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DISCUSSION

Dr. John Alverdy (Chicago, Ill.). The phenomenology of bacterial translocation is well described after hemorrhagic shock, burns, trauma, total parenteral nutrition and antibiotic administration, exploratory laparotomy, small-bowel obstruction, cyclosporine administration, neutrophil depletion, radiation enteritis, and reperfusion injury.

When we think of the victim of trauma, a picture of shock and reperfusion, blood spectrum antibiotic use, total parenteral nutrition, and inherent immunosuppression come to mind. Recent attention has focused on hypertonic saline as an ideal resuscitative fluid for the victim of trauma. Some studies have shown improvement in outcome in certain patient subsets using hypertonic saline. These investigators hypothesize that more efficient volume expansion may attenuate shock-induced edema and reperfusion injury and thus benefit the host at the brain, heart, lung, and perhaps at the gut level.

Early in my career I was taught that after stress sodium moves into the cell and drags water with it; the longer the insult, the greater the fluid shift. This edematous state, or in some patients this massive edematous state, has been viewed as innocuous. One of my teachers told me that nobody died of peripheral edema. Another one told me that he operates so quickly that sodium does not have time to move into the cell.

If hypertonic solutions reduce intestinal edema and improve gut blood flow, then prevention of the translocation of bacteriotoxic products from the gut may be beneficial to the victim of trauma. Have you convinced us that bacterial translocation was indeed decreased by the use of hypertonic saline and was this of benefit to the animals?

I have one problem with your definition of bacterial translocation. You describe positive translocation as a positive culture in either the mesenteric lymph node, liver, or spleen. Were any animals considered to be positive whose livers and spleens were positive but whose mesenteric lymph nodes were negative? And if so, how do you explain this?

One problem I have with the study design is that euvolemia is defined only by similar blood pressures between groups, which is a poor indicator of the volume resuscitation. The Wiggers' preparation has been shown over the years to result in persistent hypovolemia when the shed blood is returned after a period of shock. This may be due to a shock-induced energy deficit in the sodium potassium pump. Are we seeing that the differences between groups is merely a difference between the volume status of these groups and not the actual solutions themselves? Was urine output or oral intake measured between the groups in the 24-hour periods after operation? Do we have any other measure ensuring us that reestablishment